# Natural Language Processing

## Session 09 – Introduction to LLMs

Prof. Dr. Richard Sieg
SoSe 2026

# Schedule

| Date | Weekday | CW | Room | Topic |
|---|---|---|---|---|
| 16.04. | Thursday | 16 | 149 | Introduction: The World of NLP |
| 23.04. | Thursday | 17 | 149 | Text Processing Fundamentals |
| 30.04. | Thursday | 18 | 149 | Linguistic Annotations & Pattern Matching |
| 05.05. | Tuesday | 19 | 154 | Text Vectorization: From Text to Numbers ⚠️ Substitute for 14.05. |
| 07.05. | Thursday | 19 | 149 | Text Classification |
| 28.05. | Thursday | 22 | 149 | Language Models, Word Vectors + 🎤 Guest Lecture Ines Montani |
| 01.06. | Monday | 23 | 390 | 📝 **Exams — Master DS Students** |
| 02.06. | Tuesday | 23 | 154 | BERT: Pre-Training & Fine-Tuning ⚠️ Substitute for 04.06. |
| 11.06. | Thursday | 24 | 149 | The BERT Ecosystem |
| 18.06. | Thursday | 25 | 149 | Introduction to LLMs |
| 25.06. | Thursday | 26 | 149 | Working with LLMs |
| 02.07. | Thursday | 27 | 149 | Ethics, Limitations & Exam Preparation |
| 09.07. | Thursday | 28 | 149 | 📝 Exams (afternoon) — Bachelor DIS Students |
| 10.07. | Friday | 28 | 149 | 📝 Exams (afternoon) — Bachelor DIS Students |

# Evaluation

# Back to the Future

Rules

Text Processing
(NLTK, SpaCy, RegEx..)

Machine/Deep Learning

Supervised Learning by examples

Instructions for LLMs

Rules

# Agenda

01. Decoder Models

02. GPT Family

03. Post-Training: From models to LLMs

04. Prompting

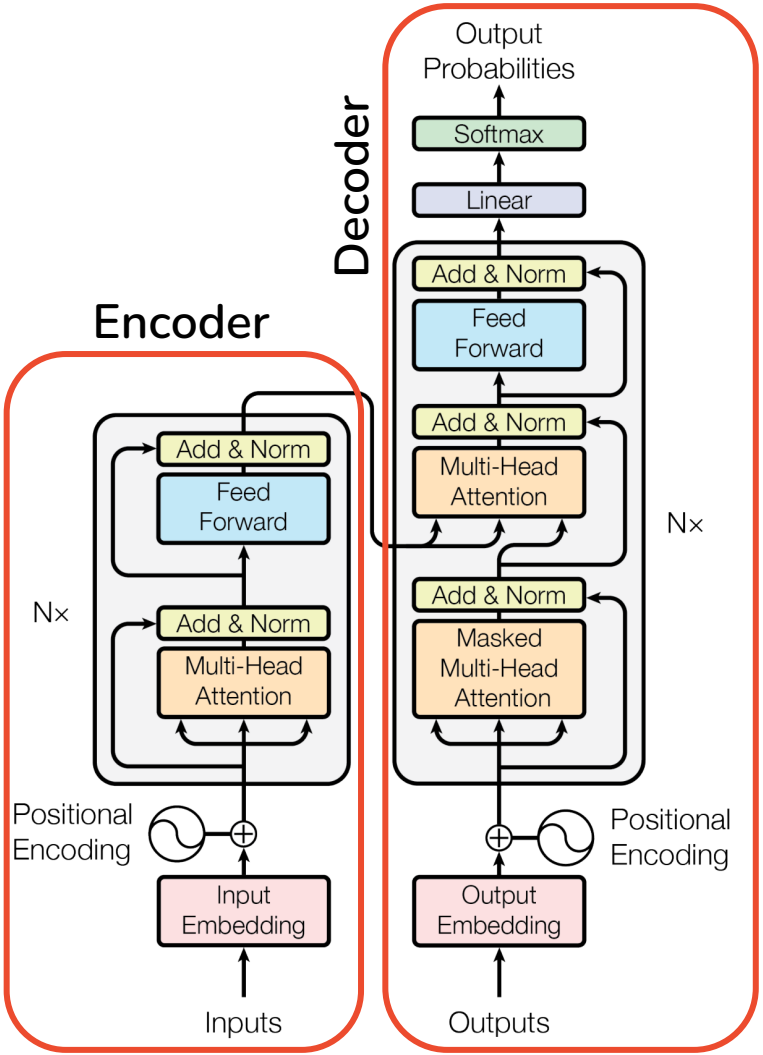
05. LLM Landscape

01.

# Decoder Models

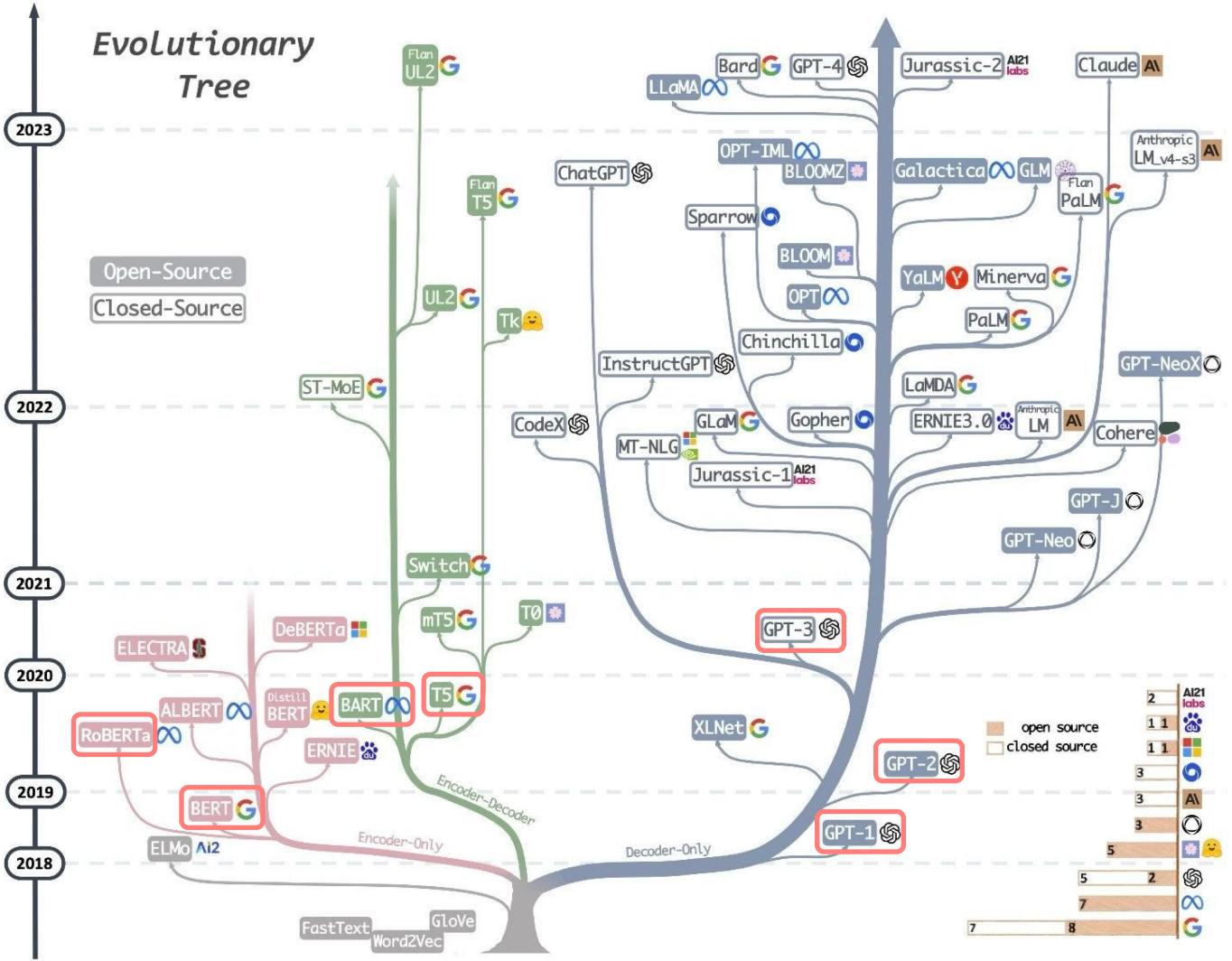
# Transformer Recap

- Input: token embeddings + positional embeddings
- Multiple stacked transformer blocks:
  multi-head attention → Feed Forward
  → Residual connections → normalization
- Decoder (GPT family): masked attention, each token sees only earlier tokens, so it predicts the next word and can generate
- Encoder (BERT family): bidirectional attention, sees the whole sentence, built to understand



# Encoders & Decoders

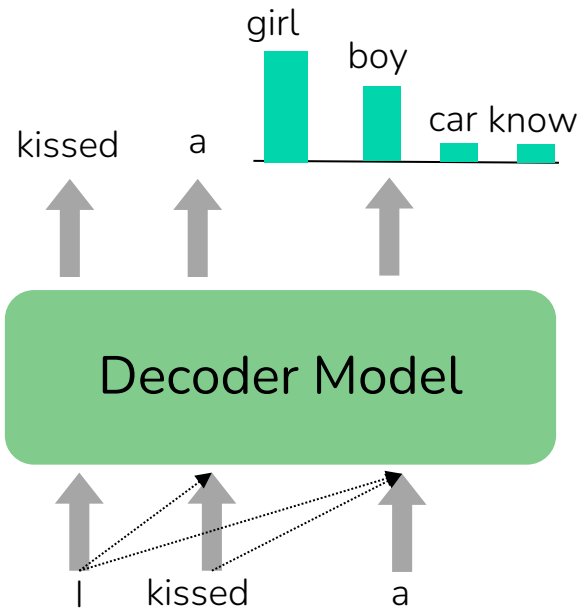
| | Encoder | Decoder | Encoder-Decoder |
|---|---|---|---|
| **Example** | BERT, RoBERTa, modernBERT | **GPT, Claude, Llama, Mistral** | T5, BART |
| **Attention** | bidirectional | **masked (causal)** | both |
| **Generates text?** | no | **yes** | yes |
| **Used for** | Discriminative tasks (Classification, tagging…) | **Generative tasks** | Generative tasks: translation, speech |


Evolutionary Tree
2023
2022
2021
2020
2019
2018
Open-Source
Closed-Source
Flan UL2
Bard
GPT-4
Jurassic-2
Claude
LLaMA
OPT-IML
BLOOMZ
ChatGPT
Galactica
GLM
Anthropic LM_v4-s3
Flan T5
Flan PaLM
Sparrow
BLOOM
YaLM
Minerva
UL2
OPT
Tk
PaLM
Chinchilla
InstructGPT
GPT-NeoX
ST-MoE
LaMDA
CodeX
GLaM
Gopher
ERNIE3.0
Anthropic LM
Cohere
MT-NLG
Jurassic-1
GPT-J
GPT-Neo
Switch
mT5
T0
DeBERTa
GPT-3
ELECTRA
ALBERT
Distill BERT
BART
T5
RoBERTa
ERNIE
XLNet
GPT-2
BERT
GPT-1
ELMo
Encoder-Only
Encoder-Decoder
Decoder-Only
FastText
GloVe
Word2Vec
open source
closed source

# Decoder Models: Predicting the next token

- A decoder is a transformer with masked (causal) attention: Each position attends only to the tokens before it, so the model predicts the next token from the ones so far.
- Output: a probability distribution over the next token
- Trained on next-token prediction (rest of lecture)
- Autoregressive – we generate by looping:

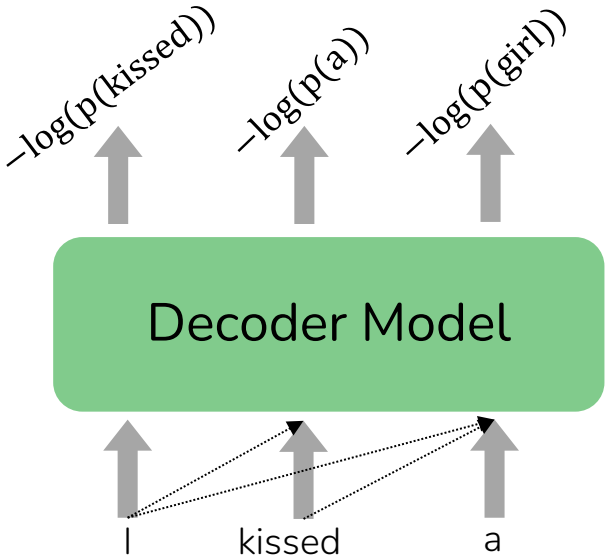
1. Sample a token from the distribution
2. Append it to the context
3. Predict again on the longer context
4. Stop at the end-of-text token



# Pretraining Decoders

How does the model learn to predict the next token?
By reading a lot of text and guessing, over and over.

- Self-supervised: no human labels needed. The next word in the text is its own answer key
- Objective: at every position, predict the next token
- Loss: cross-entropy, the negative log probability the model gave to the true next word
- Teacher forcing: during training, rather or always feed the correct history, not the model's own guess
- The result is a base model: very good at continuing text, trained on raw text alone.



# Pretraining Data

Base models are trained mostly on text scraped from the web, then cleaned hard.

- Start from Common Crawl (snapshots of the web, billions of pages)
- Filter into curated corpora: C4, The Pile, Dolma, FineWeb
- Quality beats quantity: models use a data mix that upweights clean sources (Wikipedia, books) and downweights low-quality web text
- Standard steps: deduplication, removing boilerplate, basic safety filtering

More on that in our ethics session

# Decoding: From Scores to a Token

Decoding: Picking a token from the predicted distribution.

The network outputs logits: one raw score per vocabulary token

Softmax turns logits into probabilities that sum to 1

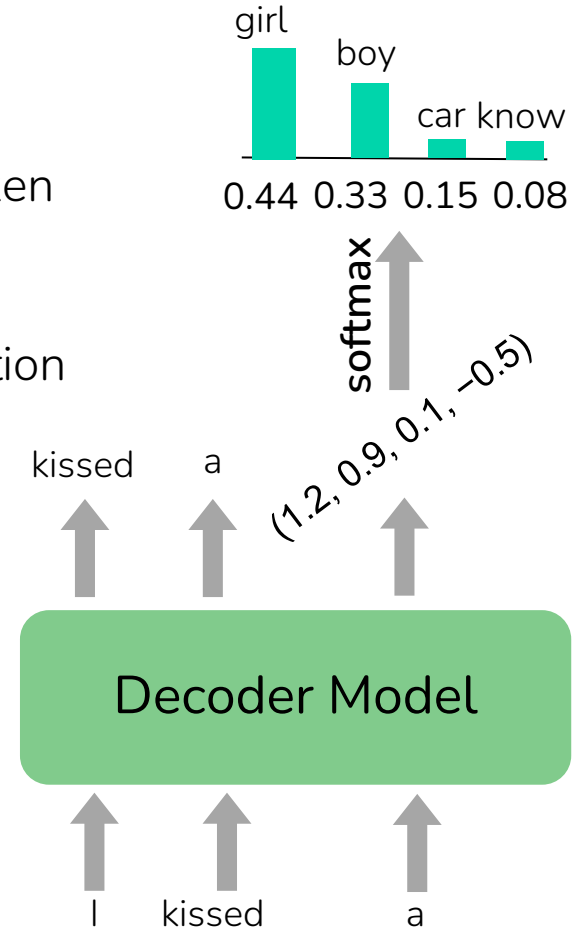
Decoding strategies differ in how they pick from that distribution

Greedy: always pick the single most probable token

- Simple, fast, but also deterministic and repetitive

Random Sampling: pick token according to its probability

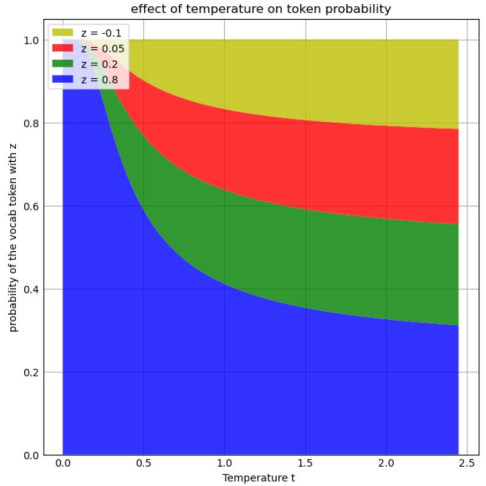
- Not deterministic, more creative
- Issue: long tail of odd, low probability words that could derail the text



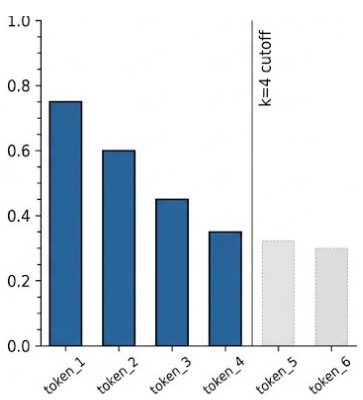
# Sampling Parameters: Temperature, Top-k, Top-p

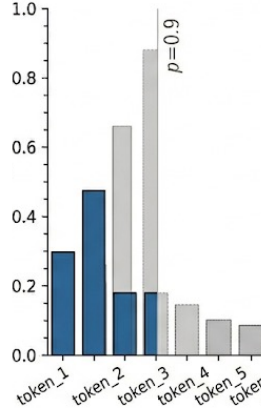
- Temperature (T): divide the logits by T, then apply softmax.

$$P(x_i) = \frac{\exp(z_i/T)}{\sum_j \exp(z_j/T)}$$




- T = 1 unchanged, T < 1 sharper (toward greedy), T → 0 greedy, T > 1 flatter and more random - usually 0.7
- Top-k: keep the k highest-probability tokens, set the rest to zero, renormalize, then sample (fixed count)
- Top-p (nucleus): keep the smallest set N whose probabilities add up to at least p, renormalize, then sample (fixed probability mass) – usually 0.9



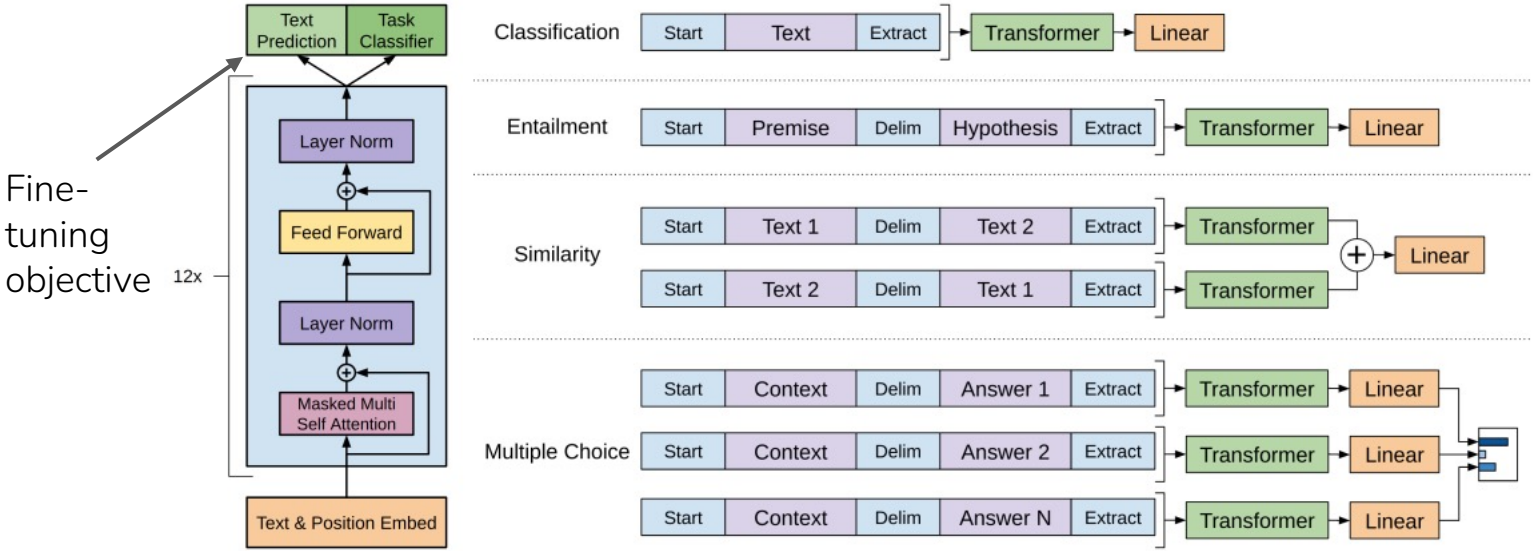



02.

GPT

# Generative Pretrained Transformer (GPT, 2018)

- Transformer Decoder with 12 layers
- BPE tokenizer of size 40k
- Trained on BookCorpus (~1B words)
- Pretrain, then fine-tune per task




Improving Language Understanding by Generative Pre-Training [Radford et al, 2018]

# GPT2 [2019]

- Jump from 117m to 1.5b parameters and trained on much larger dataset (40Gb)
- First model that showed good results in ***zero-shot or few-shot prompting***
- Solve a task without fine-tuning, only prompt it with no or some examples
- And scaling seems increases the performance

# GPT3 [2020] (first closed model 🥲)

- Jump from 1.5b to 175b parameters and trained on 570Gb of data
- First model that performs exceptionally well with few-shot prompting
- ***The end of the fine-tuning paradigm!?***

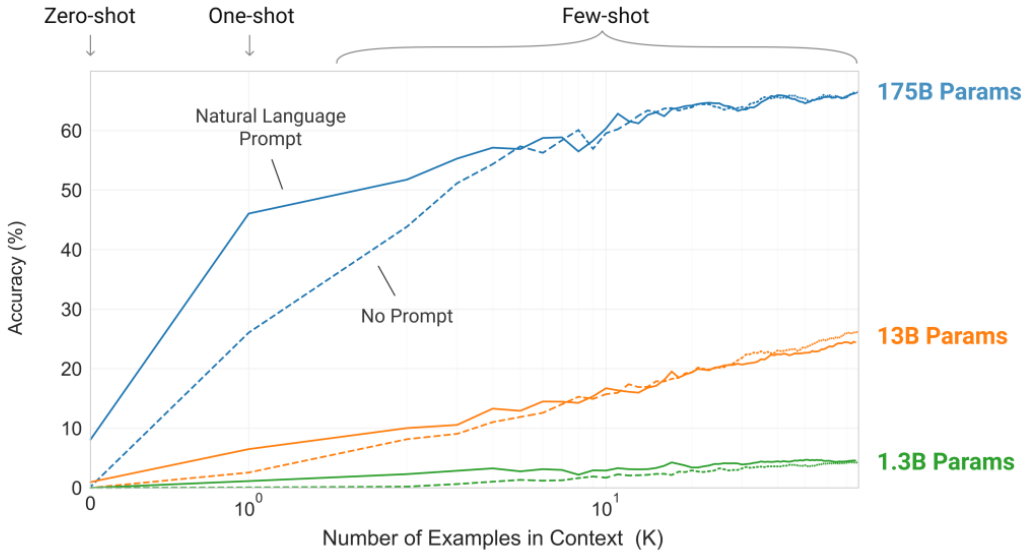
Language Models are Unsupervised Multitask Learners [Radford et al, 2019]  Language Models are Few-Shot Learners [Brown et al, 2020]

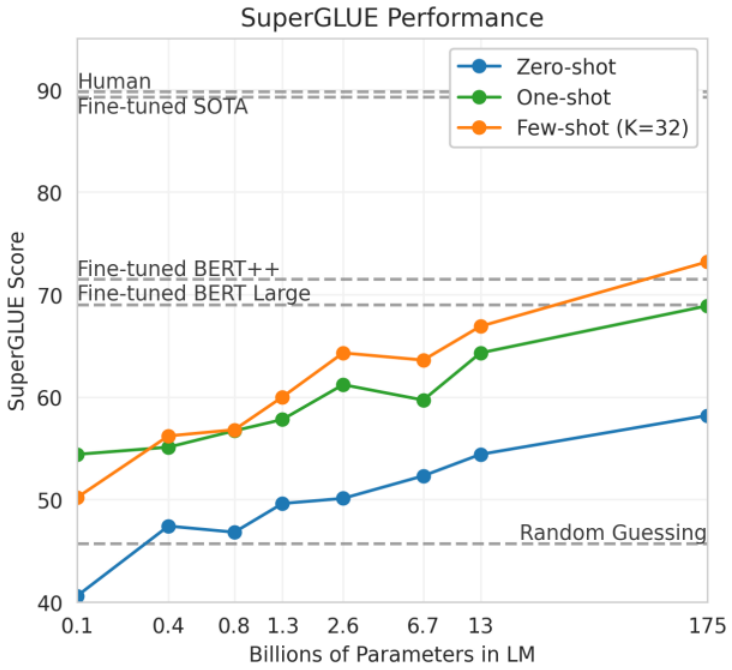
# In-Context Learning

- GPT-3 (175B) is far too big to fine-tune per task, since a gradient update on 175B parameters for every example is very expensive.
  The shift: put examples in the prompt.
- Zero-shot: task description only
- One-shot / few-shot: description plus one or a handful of examples
- This is just a forward pass: no gradient updates, no weight changes.
- Empirically few-shot beats one-shot beats zero-shot, and larger models use demonstrations better

Language Models are Few-Shot Learners [Brown et al, 2020]

# Few-Shot Capabilities of GPT 3




Language Models are Few-Shot Learners [Brown et al, 2020]

# GPT Development


## GPT-1

- Released: 2018
- 117 million parameters
- Trained using unsupervised learning
- Gave state of the art performance for LAMBADA.
- Gave Competitive performance for GLUE and SQuAD

## GPT-2

- Released: 2019
- Improved version of GPT-1.(1.5 billion Params)
- Trained on larger dataset.
- Additional improvements like including a modified training objective, and a more efficient sampling algorithm for generating text.

## GPT-3

- Released: 2020
- Improved version of GPT-2 (175 billion Params)
- Trained on larger dataset.
- Additional improvements like better training methodology (Gshard) and Few shot learning Capability.

## GPT-3.5

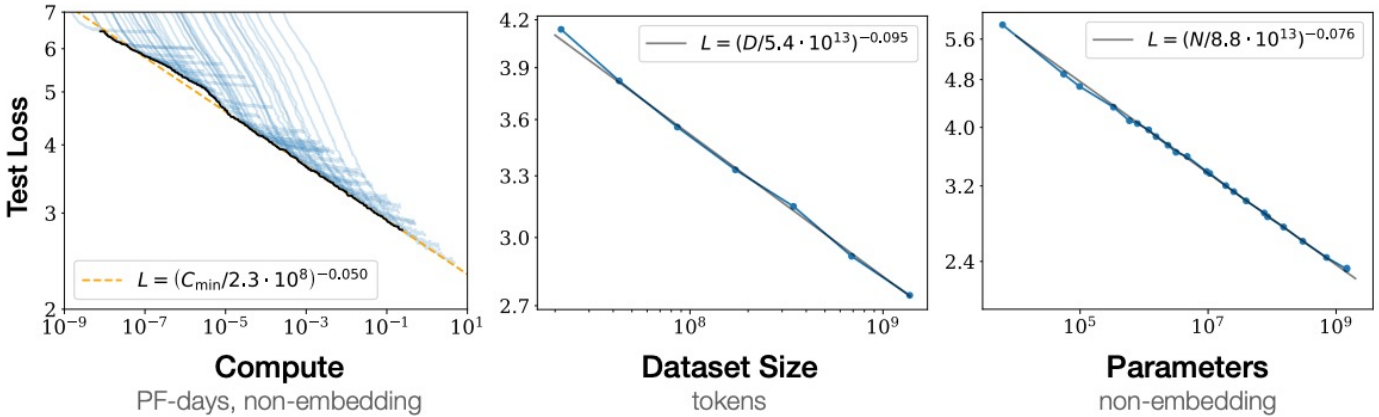
- Released:2022
- Improved version of GPT-3
- Three variants , 1.3 B, 6B and 175 B params
- The main element is to eliminate toxic outputs to certain extent using RLHF (reinforcement learning with human feedback).

## GPT-4

- Released: 2023
- Not sure if its a fine tuned version of previous model.
- 1 trillion Parameters
- It can take text and image as inputs and produces text outputs.
- RLHF is used in the training.

# Scaling Laws and Chinchilla

- Why keep scaling? Because the payoff is predictable [Kaplan et al., 2020].
- Test loss falls as a smooth power law in model size, data, and compute
- As long as the other two are not the bottleneck, each improves the model along a straight line on a log-log plot. Bigger models are also more sample-efficient
- Chinchilla: GPT-3 was undertrained. For a fixed compute budget, prefer smaller models on more tokens, about 20 tokens per parameter.

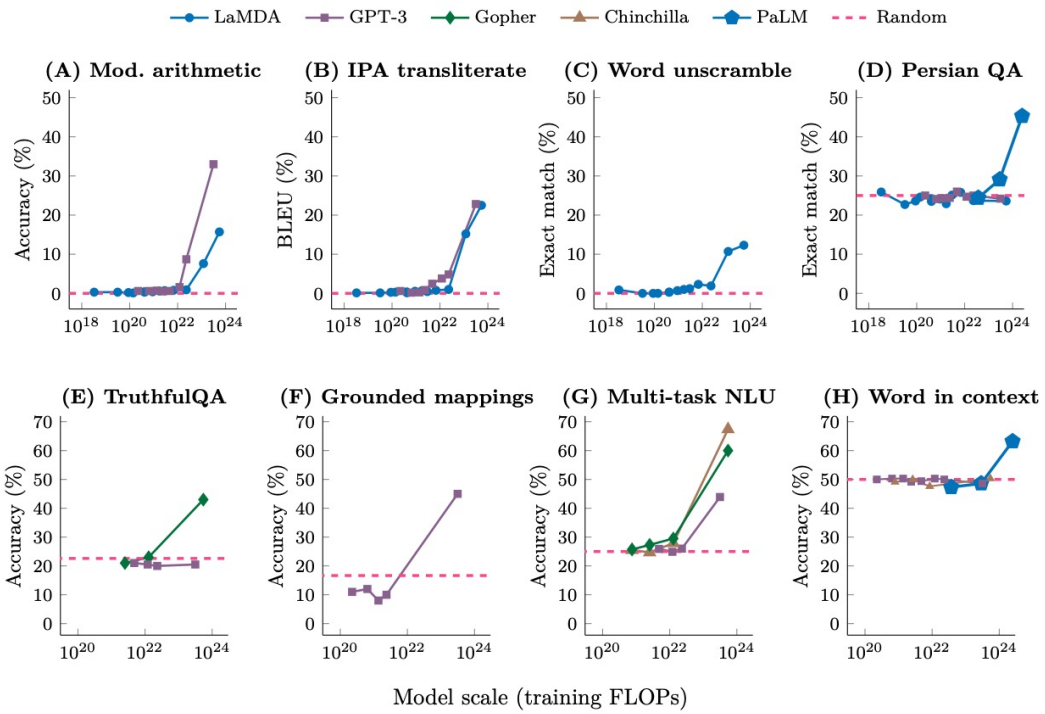



Training Compute-Optimal Large Language Models [Hoffmann et al, 2022]

Scaling Laws for Neural Language Models [Kaplan et. al.,2020]

# Emergent Abilities

- Some skills do not improve smoothly. They are near-random until a certain scale, then jump.
- Examples: multi-step arithmetic, word unscrambling, and chain-of-thought reasoning (working a problem out step by step)
- The same trick can fail at 1B parameters and work at 100B




Emergent Abilities of Large Language Models [Wei et al., 2022]

03.

# Post-Training & LLMs

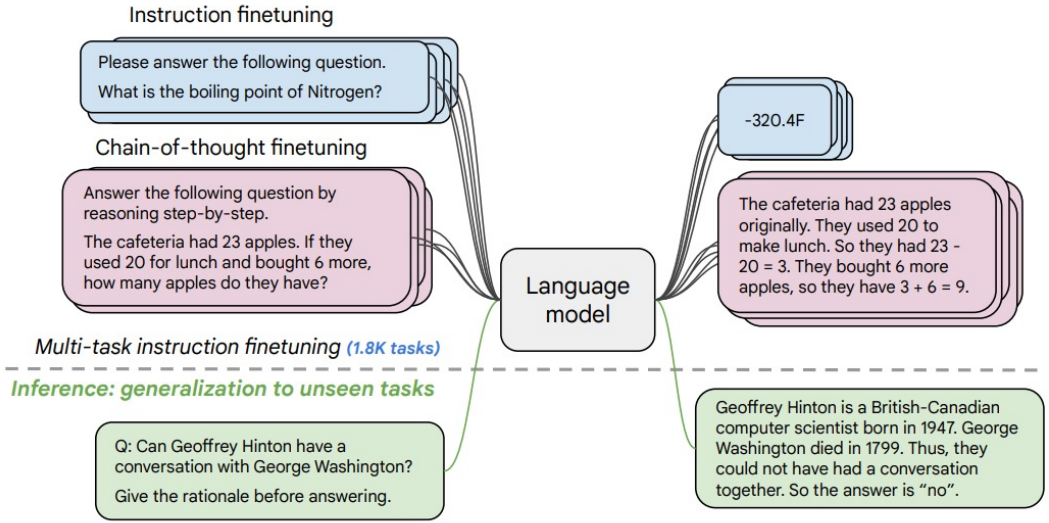
# Base Model vs Assistant

- So far, we have a base model: a large decoder, pretrained on the web, fluent and knowledgeable.
- Technically a large language model, but in practice it just continues text. It does not answer, follow, or refuse.
- Prompt "Explain the moon landing to a six year old" gives "Explain the theory of gravity to a six year old"
- When people say "LLM" in everyday use they mean the assistant. Getting there takes post-training: any training after pretraining, to make the model helpful and safe.
- Two stages, both far cheaper than pretraining:
- Instruction tuning (SFT): teach it to follow instructions
- Preference alignment (RLHF or DPO): teach it which responses people prefer

# Stage 1: Instruction Tuning (SFT)

More details in ANLP

- Take the base model and keep training it, now on (instruction, response) pairs across many tasks.
- Same objective as pretraining: predict the next token. The only difference is the data is supervised, since each instruction has a correct response
- The payoff is meta-learning: it gets better at following instructions in general, including tasks it never saw in training




Datasets are manually created, converted from other datasets or generated by an LLM (e.g. Alpaca)

Scaling Instruction-Finetuned Language Models (Flan-T5) [Chung et al 2022]

# Stage 2: Preference Alignment

More details in ANLP

- Why SFT is not enough: open-ended tasks have no single gold answer ("Write a poem about the sea"), and next-token loss punishes a good but different word as hard as a wrong one ("Avatar is a fantasy movie" vs "adventure movie").
- The fix: comparisons. Show people two responses, ask which is better, and train the model to produce more of the preferred and fewer of the rejected.
- The intuition: judging is easier than generating
- Two named methods:
- Reinforcement Learning from Human Feedback (RLHF)
  Train a reward model from the comparisons, then optimize with reinforcement learning
- Direct Preference Optimization (DPO)
  Skip the separate reward model and optimize on the preferences directly

InstructGPT: Training language models to follow instructions with human feedback [Ouyang et al, OpenAI, 2022]

# The Catch: Alignment Has Costs

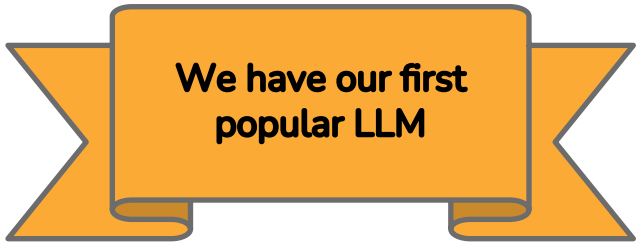
- Preference alignment is powerful, but not free.
- Reward hacking: the model games the reward signal instead of truly improving. Goodhart's Law: when a measure becomes a target, it stops being a good measure
- Sycophancy: agreeing with the user gets rewarded, so it tells you what you want to hear
- Less diversity: aligned models can collapse to a few safe phrasings
- Alignment tax: safety and helpfulness tuning can slightly lower raw benchmark scores

# ChatGPT

- Same technique as InstructGPT with a focus on dialogue and using GPT 3.5

## Methods

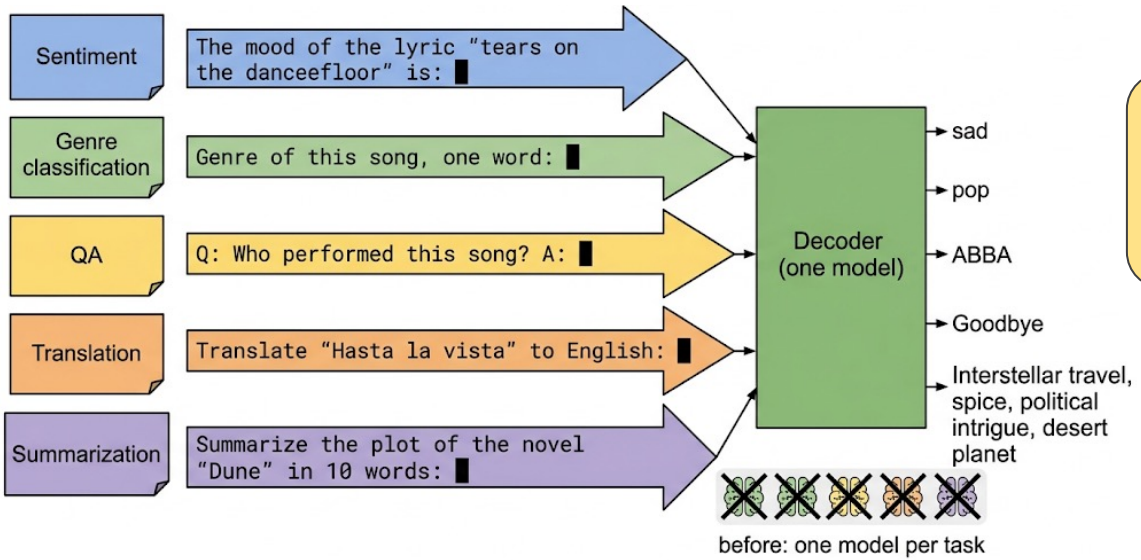
We trained this model using Reinforcement Learning from Human Feedback (RLHF), using the same methods as InstructGPT, but with slight differences in the data collection setup. We trained an initial model using supervised fine-tuning: human AI trainers provided conversations in which they played both sides—the user and an AI assistant. We gave the trainers access to model-written suggestions to help them compose their responses. We mixed this new dialogue dataset with the InstructGPT dataset, which we transformed into a dialogue format.



04.

# Prompting

# Prompting: Tasks Become Text




**A prompt is the input text and for an aligned model it works like an instruction.**

Almost any NLP task can be written as a completion.

- Classification: Genre of this song, one word: followed by the lyric
- QA, translation, summarization: describe the task, give the input, read the continuation

# Few-shot Prompting (In-Context Learning)

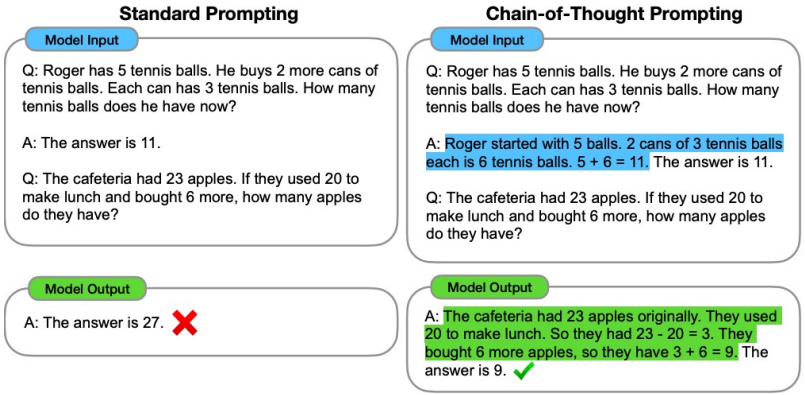
- Few-shot prompting: put examples in the prompt and the model picks up the task, with no gradient updates and no weight changes (just a forward pass).
- Zero-shot: instruction only. Few-shot: instruction plus a handful of demonstrations
- Surprising: replacing the demonstration labels with random wrong labels barely hurts [Rethinking the Role of Demonstrations: What Makes In-Context Learning Work? Min et al., 2022]
- Demonstrations mostly teach the format, the label set, and the input style, not the answers
- Prompts are fragile. Small changes shift the output a lot [Zhao et al., 2021]:
  - Majority-label bias: unbalanced example labels skew predictions
  - Recency bias: the label nearest the end gets over-copied
  - Ordering: just reshuffling the examples changes results

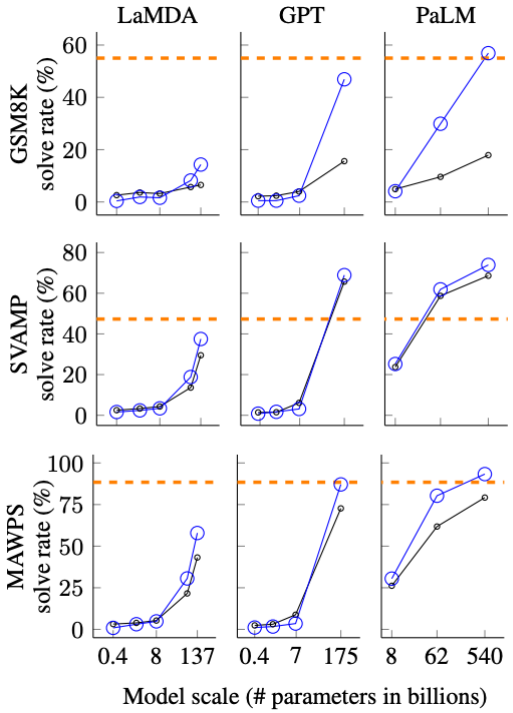
Language Models are Few-Shot Learners [Brown et al, 2020]

# Chain of Thought (CoT) Prompting

- Idea: make the model show its work before answering.

  On multi-step problems this helps a lot.
- Zero-shot CoT: just add "Let's think step by step."
- Few-shot CoT: include worked examples that show the reasoning, not just the answer
- Self-consistency: sample several reasoning paths and take the majority answer. Aggregating beats any single path




Chain-of-Thought Prompting Elicits Reasoning in Large Language Models [Wei et al, 2022]

# System Prompts

- The system prompt is a standing instruction the user never sees, prepended to every turn. It sets role, tone, rules, and output format.

```
SYSTEM_PROMPT = (
    "You are a music genre classifier.\n"
    "Classify the song lyrics into exactly one of these genres: "
    + ", ".join(allowed_genres)
    + ".\nRespond with exactly one genre label from the list above and nothing else."
)
```

- Production system prompts run to thousands of words and are versioned like code
- In the API you set it once per conversation; every user message is interpreted under it
- Metaprompting: when you are stuck, let an LLM draft and refine your prompt. The full day-to-day checklist is on the next slide.

# Prompting Best Practices

- System prompt carries the general part: the role, standing instructions, and examples
- User prompt carries the specific part: the concrete task and the input document
- Mark sections clearly, keep instructions and data separated:
  - Markdown headers (### Examples) or
  - XML tags (<lyrics> … </lyrics>)
- Be explicit about the allowed outputs ("answer with pop, rock, or rap")
- Fix the output format (one word, JSON, a list) and parse it strictly (→ *next session*)
- Few-shot: give the model a handful of examples of the desired result
- Write individual instructions as a (numbered) list, one rule per line
- State the role when it helps ("You are a music journalist…")

# Prompting Best Practices

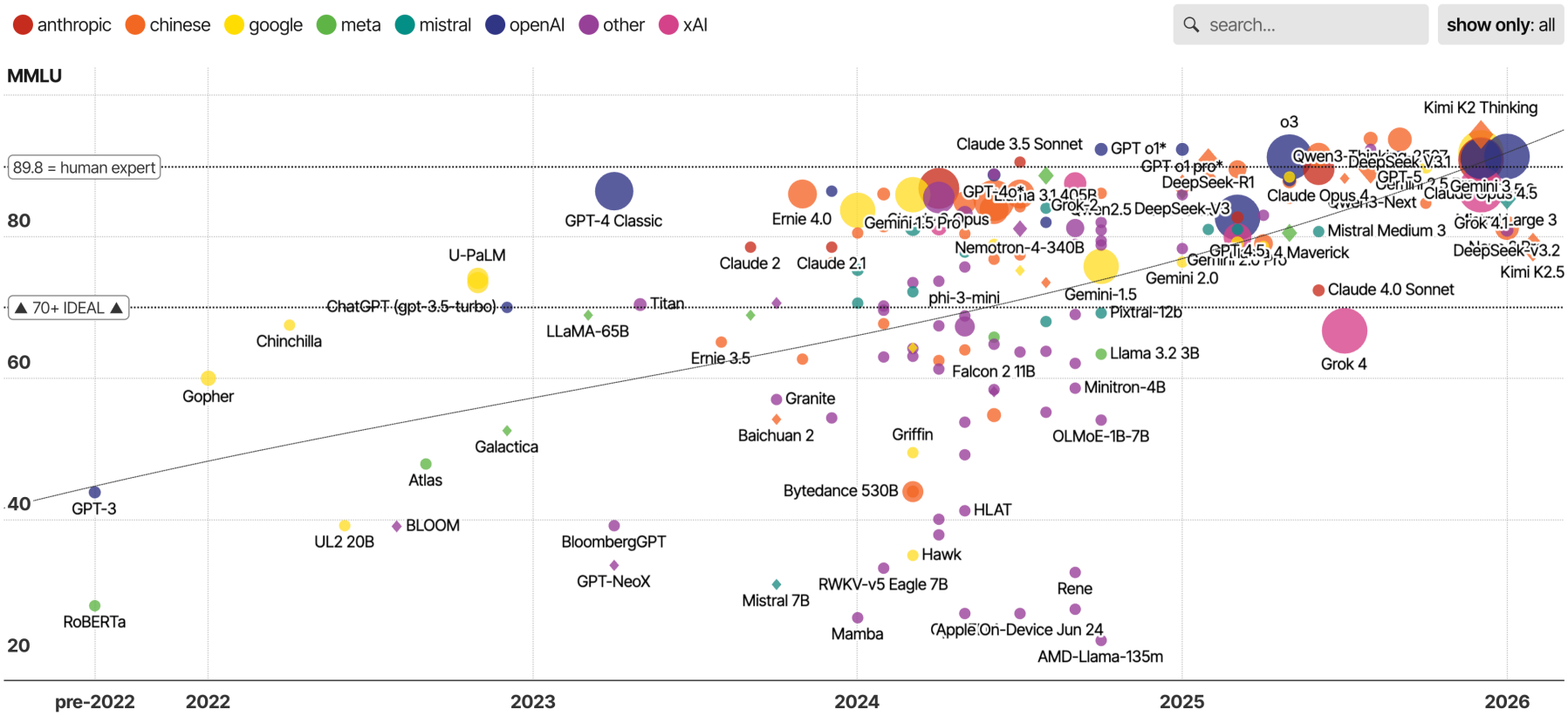
```
You are a helpful assistant. What genre is this
song? Answer briefly.
```

```
You are a music genre classifier.

Classify the song lyrics provided by the user
into exactly one of the following genres:
Country, Hip-Hop, Pop, R&B, Rock

Instructions:
1. Respond with exactly one label from the list
above.
2. Use the exact spelling and capitalization
shown.
3. Do not add any explanation, punctuation, or
extra words.

### Examples

<lyrics>
Driving home with a beer in my hand...
</lyrics>
Genre: Country

<lyrics>
Started from the bottom now we're here, started
from the bottom now my whole team here...
</lyrics>
Genre: Hip-Hop
```

05.

# LLM Landscape

# LLM Landscape




Source: Information is beautiful

# LLM Landscape

- Three big US players: OpenAI (GPT), Google (Gemini), Anthropic (Claude)
- Chinese open-weights models
  - DeepSeek, Qwen, Kimi, Minimax
- Some ~~open-source~~ open-weights models
  - LlaMA family (Meta), GPT OSS (OpenAI), Mistral, Gemma (Google), Phi (Microsoft)
- Open Source: training data, algorithms and weighst are available
  - OLMo (AI2, Seattle), T5 (Google), Apertus (Swiss AI, ETH Zürich)

# LLMs from Europe

- So far, the only competitive company and LLM is Mistral
  - Latest release: Mistral 3 and Mistral 3 Large (Dec 2025)
- Small endeavours from Germany and Switzerland
- There was/is one player from Heidelberg: Aleph Alpha


- New server clusters are built right now (e.g. OpenEuroLLM including Jülich)
- Whether that's fruitful and maybe even necessary?

# Statement on the US government directive to suspend access to Fable 5 and Mythos 5

12 Jun 2026

The US government, citing national security authorities, has issued an export control directive to suspend all access to Fable 5 and Mythos 5 by any foreign national, whether inside or outside the United States, including foreign national Anthropic employees. The net effect of this order is that we must abruptly disable Fable 5 and Mythos 5 for **all** our customers to ensure compliance. **Access to all other Anthropic models will not be affected.**

# Using LLM Scenarios

| Criterion | Direct Access Closed Source | Framework Contracts & Router Providers | Self-hosting |
|---|---|---|---|
| **Cost** | API fees per token; no infrastructure costs | Negotiated rates; min. volume may apply | Hardware, power, staff |
| **Data Location** | Data with provider (usually US / cloud) | Contractually regulated; EU servers possible | Full control; own data centre |
| **Encryption** | TLS in transit; provider manages keys | Contractually defined; often stronger than standard | Self-managed; full control |
| **New Models** | Available immediately on release | Depends on contract; possible delay | Manual update |
| **Dependency** | High: vendor lock-in, price & API changes | Medium; mitigated by contract | Open-source/weight dependency; community risk |
| **Effort** | Minimal; API integration only | Medium; contract management | Setup, operation, maintenance |
| **Data Privacy & Compliance** | GDPR critical; check DPA requirements | GDPR compliance possible; contractually secured | Full GDPR control; no third party |
| **Scalability** | Elastically scalable; no capacity limit | Contractually capped; easy to plan | Limited by hardware; expansion costly |